Small Sample Tests

Student's t tests, the correlation test, chi-square tests and the F test for variances.

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1 Small samples

Often $n < 30$ (not a strict boundary). Uses the exact t, χ^2 and F distributions.

W.S. Gosset published the t distribution as "**Student**". It has heavier tails than the normal.

2 Student's t tests for means

Test	Statistic	df
One sample, $H_0: \mu = \mu_0$	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$n - 1$
Two samples, equal variances	$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$, pooled $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$n_1 + n_2 - 2$
Welch (unequal variances)	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	Welch–Satterthwaite (not an integer)
Paired: before/after on the same unit	$t = \frac{\bar{d}}{s_d/\sqrt{n}}$, $d_i = x_i - y_i$	$n - 1$

3 Testing a correlation

$H_0: \rho = 0$, **df** $= n - 2$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Assumes a bivariate normal model.

4 F test: two variances

$H_0: \sigma_1^2 = \sigma_2^2$

$$F = \frac{s_1^2}{s_2^2}, \quad df = (n_1 - 1,\ n_2 - 1)$$

Put the larger variance on top ($F \geq 1$) and use the upper $\alpha/2$ cut-off for a two-sided test.

5 Chi-square tests

Test	Statistic	df
Goodness of fit	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$	$n - 1 - m$
Independence ($r \times s$ table)	$\chi^2 = \sum\sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$, $E_{ij} = \frac{A_i B_j}{N}$	$(r-1)(s-1)$
2 × 2 table	$\chi^2 = \frac{n(ad-bc)^2}{C_1 C_2 R_1 R_2}$	1
Population variance	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$n - 1$

n categories, m parameters estimated from the data. Expected = row total × column total ÷ grand total. Goodness-of-fit and independence use the **upper** tail.

6 Yates' correction

Optional continuity adjustment for 2 × 2 tables:

Corrected

$$\frac{n[\max(0, |ad - bc| - \frac{n}{2})]^2}{C_1 C_2 R_1 R_2}$$

With small expected counts, consider an exact test.

7 Which test?

Question	Test
Does mean plant height equal 90 cm?	One-sample t
Do two independent varieties differ in yield?	Two-sample t (pooled or Welch)
Did yield change on the same plots before and after a treatment?	Paired t
Do segregation counts fit a 9:3:3:1 ratio?	χ^2 goodness of fit
Is adoption independent of education level?	χ^2 independence
Are two varieties equally variable?	F test

! Common mistakes

- × Two-sample t on paired data.
- × Wrong df: $n - 1$ vs $n_1 + n_2 - 2$ vs $(r-1)(s-1)$.
- × Running χ^2 on percentages instead of counts.

? Fill in the blanks

1. A one-sample t test on 12 plots has ____ degrees of freedom. [11]
2. Two independent groups of 8 and 10 plots (equal variances) give ____ degrees of freedom. [16]
3. Yields of the same 15 plots before and after mulching are compared with the ____ t test. [paired]
4. A 3 × 4 contingency table has ____ degrees of freedom. [6]
5. Row total 40, column total 30 and grand total 100 give an expected frequency of ____. [12]
6. Sample variances 18 and 6 give F = ____. [3]

Source: *Textbook of Agricultural Statistics*, Ch. 15 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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